Quant Interview Preparation Guide

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Contents

1 Day-Before Priorities (1-page) **2**

2 Introduction **3**

3 Probability **4**
- 3.1 Key Concepts . . . 4
- 3.2 Standard Distributions . . . 5

4 Linear Algebra **7**
- 4.1 Matrix Decompositions . . . 7
- 4.2 Key Properties . . . 7
- 4.3 Named Formulas & Theorems . . . 8

5 Statistics **9**
- 5.1 Estimation Theory . . . 9
- 5.2 Hypothesis Testing . . . 9
- 5.3 ANOVA (Analysis of Variance) . . . 10

6 Regression **11**
- 6.1 OLS . . . 11
- 6.2 Assumptions (Gauss-Markov) . . . 11
- 6.3 Metrics . . . 11

7 Python for Quants **13**
- 7.1 Data Structures . . . 13
- 7.2 Key Libraries . . . 13

8 Finance Concepts **14**

9 Time Series **15**
- 9.1 Core Concepts . . . 15

10 Special Topics **16**
- 10.1 Optimization with Costs . . . 16
- 10.2 Microstructure & Execution . . . 16
- 10.3 Whiteboard Cheats (full step outlines) . . . 16

10.4 When in Doubt (stuck playbook) 17

1 Day-Before Priorities (1-page)

Use this as the front sheet. Hit these in order, time-boxing to what you can finish.

1. **Tell 3 resume stories** (execution cost win, signal discovery, turnover paper) with numbers, failure/lesson variant ready.
2. **Intraday alpha + execution**: How you build/validate intraday signals (features, decay, leakage checks), and how you size/route trades (arrival vs VWAP/POV choice, slippage breakdown).
3. **Time-series core**: AR(1) stationarity/half-life, ADF null, cointegration test steps, GARCH vs EWMA.
4. **Optimization with costs**: Mean-Var + turnover form, KKT shape, L1 vs L2 choice, Woodbury use case.
5. **Execution framing**: slippage decomposition, arrival vs VWAP vs POV why/when, queue/selection risk.
6. **Brain-teaser cadence**: one probability puzzle warm-up (from your list) to get into flow.
7. **Python drills**: vectorized rolling z-score, groupby+merge, AR(1) sim/phi-hat, MC GBM, toy order-book PnL.
8. **Ask-back questions**: have 2 per interviewer type (PM: risk budget/process; researcher: validation/overfit controls).

2 Introduction

This document serves as a comprehensive guide for quantitative interview preparation, covering probability, linear algebra, statistics, regression, Python, and finance concepts.

3 Probability

Intuition

Probability is the logic of uncertainty. In quant finance, it is the bedrock of modeling asset prices and risk. We deal with random variables, expectations, and conditional probabilities to quantify the likelihood of future events given current information (filtration). Key focus: Bayes' rule for updating beliefs and distributions for modeling returns.

Key points

- Linearity of expectation + indicators are your fastest tools.
- Conditioning/Bayes is the right lens when information arrives (prices, fills).
- Check tails: know when Normal fails; remember exponential/Poisson memorylessness.
- For puzzles, turn counting into geometry (areas) or permutation cycles (prisoners).

Motivating Example

The Monty Hall Problem:
Setup: You are on a game show. There are 3 doors. Behind one is a Car; behind the others, Goats. You pick a door (say, Door 1). The Host, who knows what is behind the doors, opens one of the other doors (say, Door 3) which has a Goat. He then asks: "Do you want to switch to Door 2?"
Intuition: Yes. Your initial probability was 1/3. The "Rest of World" had 2/3. Host opening a door doesn't change your 1/3, but concentrates the "Rest of World" 2/3 probability entirely onto Door 2.
Bayesian Solution: Let C_i denote event "Car is behind Door i". Let H_3 denote "Host opens Door 3".
We want $P(C_2|H_3)$ vs $P(C_1|H_3)$.

$$P(C_1|H_3) \propto P(H_3|C_1)P(C_1) = \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{6}$$
$$P(C_2|H_3) \propto P(H_3|C_2)P(C_2) = 1 \cdot \frac{1}{3} = \frac{1}{3}$$

Sum of weights $= 1/6 + 1/3 = 1/2$. Normalizing:

$$P(C_2|H_3) = \frac{1/3}{1/2} = \frac{2}{3} \approx 66.7\%.$$

Verdict: **Switch**.

3.1 Key Concepts

- **Bayes' Theorem**: $P(A|B) = \frac{P(B|A)P(A)}{P(B)}$
- **Expected Value**: $\mathbb{E}[X] = \sum xp(x)$ or $\int xf(x)dx$
- **Variance**: $\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$
- **Linearity of Expectation**: $\mathbb{E}[X+Y] = \mathbb{E}[X] + \mathbb{E}[Y]$ (always, even if dependent!)

- **Indicator Functions**: $\mathbf{1}_A = \begin{cases} 1 & \text{if } A \text{ occurs} \\ 0 & \text{otherwise} \end{cases}$. Key property: $\mathbb{E}[\mathbf{1}_A] = P(A)$.

  Power move: Write count as sum of indicators, then use linearity. E.g., $X = \sum_i \mathbf{1}_{A_i} \implies \mathbb{E}[X] = \sum_i P(A_i)$.

- **Markov Chain Return Time**: For an ergodic chain with stationary distribution π, the expected return time to state i is $\mathbb{E}[T_{ii}] = 1/\pi_i$.

3.2 Standard Distributions

Bernoulli(p) – Single trial

PMF: $P(X = k) = p^k(1-p)^{1-k}, \quad k \in \{0, 1\}$ **Mean**: p **Var**: $p(1-p)$

Ex: A trade has 60% chance of profit. Expected profitable trades in 1 trial? **Sol**: $\mathbb{E}[X] = 0.6$

Binomial(n and p) – Sum of n Bernoullis

PMF: $P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$ **Mean**: np **Var**: $np(1-p)$

Ex: 20 bonds, each 5% default prob. P(exactly 2 defaults)?
Sol: $\binom{20}{2}(0.05)^2(0.95)^{18} = 190 \times 0.0025 \times 0.397 \approx 18.9\%$

Geometric(p) – Trials until first success

PMF: $P(X = k) = (1-p)^{k-1}p, \quad k \geq 1$ **Mean**: $1/p$ **Var**: $(1-p)/p^2$
Memoryless: $P(X > m + n \mid X > m) = P(X > n)$

Ex: Strategy wins 25% of days. Expected days until first win? **Sol**: $1/0.25 = 4$ days

Poisson(lambda) – Rare event counts

PMF: $P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad k \geq 0$ **Mean** = **Var** = λ

Ex: Orders arrive at 10/min. P(exactly 15 in a minute)?
Sol: $\frac{10^{15}e^{-10}}{15!} \approx 3.5\%$

Exponential(lambda) – Time between Poisson events

PDF: $f(x) = \lambda e^{-\lambda x}, \quad x \geq 0$ **Mean**: $1/\lambda$ **Var**: $1/\lambda^2$
Memoryless: $P(X > s + t \mid X > s) = P(X > t)$ **Survival**: $P(X > x) = e^{-\lambda x}$

Ex: Avg 2 sec between trades ($\lambda = 0.5$). P(wait > 3s)? **Sol**: $e^{-1.5} \approx 22.3\%$

Normal(mu and sigma squared) – CLT limit

PDF: $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$ **Mean**: μ **Var**: σ^2
Quantiles: $z_{0.90} = 1.28$, $z_{0.95} = 1.645$, $z_{0.975} = 1.96$, $z_{0.99} = 2.33$

Ex: Returns $\sim N(0.05\%, 2\%)$. 5% downside VaR?
Sol: $\mu - z_{0.95}\sigma = 0.05\% - 1.645 \times 2\% = -3.24\%$

LogNormal – Product of positive shocks

If $\ln X \sim N(\mu, \sigma^2)$: **Mean**: $e^{\mu+\sigma^2/2}$ **Var**: $e^{2\mu+\sigma^2}(e^{\sigma^2} - 1)$

Ex: $S_0 = \$100$, log-returns $\sim N(0, 0.04)$. Expected price in 1 year?
Sol: $100 \times e^{0+0.02} \approx \$102.02$

Pareto / Power Law – Fat tails

PDF: $f(x) = \frac{\alpha x_m^\alpha}{x^{\alpha+1}}$, $x \geq x_m$ **Mean**: $\frac{\alpha x_m}{\alpha-1}$ (if $\alpha > 1$)
Survival: $P(X > x) = (x_m/x)^\alpha$ Variance $\to \infty$ if $\alpha \leq 2$

Ex: Losses Pareto with $\alpha = 2.5$, $x_m = \$1M$. $P(\text{loss} > \$2M)$?
Sol: $(1/2)^{2.5} = 1/\sqrt{32} \approx 17.7\%$

Practice problems for this section have been moved to `problem_bank/probability/` *and* `problem_bank/brain_teas`

4 Linear Algebra

Intuition

Linear Algebra is the study of flat geometry. In our context, we distinguish rigorously between **vectors** (allocations w^i) and **covectors** (alphas α_i). The metric tensor g_{ij} (covariance Σ) maps between them and defines distances/angles. Operations like $w^T \Sigma w$ are better understood as contractions $\Sigma_{ij} w^i w^j$ or inner products $\langle w, w \rangle_\Sigma$.

Key points

- Spectral view clarifies risk: eigenvectors = uncorrelated portfolios; eigenvalues = risk along them.
- Conditioning matters: ill-conditioned $X^T X$ inflates standard errors; fix with regularization or basis change.
- PSD checks (determinants, eigenvalues, Schur complement) stop you from using invalid covariances.
- Rank-1/low-rank updates (Sherman–Morrison/Woodbury) are must-know for fast rebalances.

4.1 Matrix Decompositions

- **Eigenvalue Decomposition** ($A = Q\Lambda Q^{-1}$): Diagonalizes the matrix. Transforms coupled systems into uncoupled ones. Key for analyzing stability and powers of matrices.
- **SVD** ($A = U\Sigma V^T$): The fundamental theorem of linear algebra. Applicable to *any* matrix (even non-square). Used in PCA (dimension reduction) and solving ill-posed linear least squares.
- **QR Decomposition** ($A = QR$): Orthogonal Q times upper triangular R. Numerically stable for solving $Ax = b$ (back-substitution on $Rx = Q^T b$). Foundation of Gram-Schmidt orthogonalization.
- **Cholesky** ($A = LL^T$): "Square root" of a PSD matrix. Crucial in finance for generating correlated random variables ($z_{corr} = Lz_{indep}$) for Monte Carlo.

4.2 Key Properties

- **Rank**: Dimension of the column space. The number of linearly independent bets you have.
- **Determinant**: Product of eigenvalues. Represents volume scaling factor. If 0, the matrix collapses volume (non-invertible).
- **Trace**: Sum of eigenvalues. Invariant under change of basis.
- **Positive Definite**: $x^T A x > 0$ for all $x \neq 0$. Represents a valid covariance structure (variance must be positive).

4.3 Named Formulas & Theorems

- **Cramer's Rule**: For $Ax = b$ where A is $n \times n$, $x_i = \frac{\det(A_i)}{\det(A)}$ where A_i is A with column i replaced by b.
- **Sherman-Morrison**: $(A + uv^T)^{-1} = A^{-1} - \frac{A^{-1}uv^TA^{-1}}{1+v^TA^{-1}u}$. Rank-1 update to inverse.
- **Perron-Frobenius**: For a matrix with positive entries, the largest eigenvalue is real, positive, and unique. For stochastic matrices, $\lambda_{max} = 1$.

Practice problems for this section have been moved to `problem_bank/linear_algebra/`.

5 Statistics

Intuition

While Probability predicts the future from a known model, Statistics infers the model from past data. We never know the true parameters (μ, σ), so we build estimators. The core challenge is distinguishing signal from noise (inference), avoiding overfitting, and understanding the stability of our estimates (confidence intervals, hypothesis testing).

Key points

- Always ask: bias vs variance vs sample size; CI width scales as $1/\sqrt{n}$.
- Dependence kills naive p-values—use HAC/Newey–West or bootstrap if residuals autocorrelate.
- Multiple testing: control false discoveries (FDR) if you scan many signals.
- For backtests, focus on out-of-sample decay and robustness, not just in-sample Sharpe.

Motivating Example

Market Efficiency Test: You believe a strategy has a Sharpe Ratio of 1.0. You run it for 1 year (252 days) and get $\hat{SR} = 0.5$. Is the strategy broken?
Standard Error of Sharpe is $\approx \sqrt{1/T}$. Here $T = 1$ (year), so $SE \approx 1$.
$Z = \frac{0.5-1.0}{1} = -0.5$.
A Z-score of -0.5 is very common (p-value $\approx$ 30%). You cannot reject the hypothesis that the strategy is fine. You simply had a bad year (noise).

5.1 Estimation Theory

- **Central Limit Theorem (CLT)**: Why do we obsess over Normal distributions? because the average of *any* i.i.d. noise (with finite variance) converges to it. It justifies OLS and mean-variance optimization.
- **Law of Large Numbers (LLN)**: Averages converge to Expectations. This justifies using historical data to estimate future parameters ("Ergodicity").
- **MLE (Maximum Likelihood)**: The general framework for parameter estimation. Find θ that maximizes the probability of seeing the data we actually saw.
  $\hat{\theta}_{MLE} = \text{argmax}_\theta \prod f(x_i; \theta)$. Often biased! (e.g., variance).
- **Bias-Variance Tradeoff**:
  MSE = Bias^2 + Variance.
  Complex models (low bias) fit noise (high variance). Simple models (high bias) miss signal (low variance).

5.2 Hypothesis Testing

- **Z/t tests**: Means with known/unknown σ. Use t with $n-1$ dof when σ estimated. $H_0 : \mu = \mu_0$; statistic $(\bar{x} - \mu_0)/(s/\sqrt{n})$.

- χ^2: Variance test for Normal data: $(n-1)s^2/\sigma_0^2 \sim \chi^2_{n-1}$.
- **F-test**: Ratio of variances or joint significance in regression (compare restricted vs full).
- **Likelihood ratio vs Wald vs Score**: Three equivalent large-sample tests; $\text{LR} = -2(\log L_{restr} - \log L_{full}) \sim \chi^2_k$.
- **Power**: $1-\beta$; increases with effect size, sample size, lower noise.
- **Multiple testing/FDR**: Benjamini–Hochberg sorts p-values and controls expected false discovery proportion; use when scanning alphas.
- **Bootstrap**: Resample to get empirical CI/p-values when assumptions fail (heavy tails, dependence); block bootstrap for time series.

5.3 ANOVA (Analysis of Variance)

- **Purpose**: Test whether means across k groups are equal. $H_0 : \mu_1 = \mu_2 = \cdots = \mu_k$.
- **Key Idea**: Decompose total variance into *between-group* (explained) and *within-group* (residual) components.
$$SS_{Total} = SS_{Between} + SS_{Within}$$
- **F-statistic**:
$$F = \frac{MS_{Between}}{MS_{Within}} = \frac{SS_{Between}/(k-1)}{SS_{Within}/(N-k)} \sim F_{k-1,N-k}$$
Large $F \Rightarrow$ group means differ more than expected by chance.
- **Assumptions**: Independence, Normality within groups, Homoscedasticity (equal variances). Robust to mild violations with large n.
- **One-way vs Two-way**: One-way tests one factor; Two-way tests two factors and their interaction.
- **Post-hoc tests**: ANOVA says "some means differ"; use Tukey HSD, Bonferroni, or Scheffé to identify *which* pairs differ (controls family-wise error).

Practice problems for this section have been moved to `problem_bank/statistics/`.

6 Regression

Intuition

Regression is the workhorse of alpha generation. It quantifies relationships between variables (e.g., Returns vs Factors). OLS projects the dependent variable onto the subspace spanned by independent variables. We look for β (sensitivity/edge) and ϵ (residual risk). Understanding assumptions (Gauss-Markov) reveals when our models fail (heteroskedasticity, serial correlation).

Motivating Example

CAPM: $r_{AAPL} - r_f = \alpha + \beta(r_{Mkt} - r_f) + \epsilon$.
We regress Apple excess returns on Market excess returns.
If $\beta = 1.2$, Apple is 20% more volatile than the market.
If $\alpha > 0$ (statistically significant), Apple generated returns *beyond* what its risk exposure explains.
As quants, we hunt for predictors X that generate predictors of α.

Key points

- Interpret β as projection: OLS = orthogonal projection; R^2 is fraction of variance captured.
- Watch assumptions: autocorr/heteroskedasticity break naive SEs—use HAC/Newey–West.
- Omitted variable bias = correlation between regressors; check direction via $\delta_{21}\beta_2$ term.
- Regularization choice: L1 for sparsity/turnover control; L2 for stability/collinearity.

6.1 OLS

Model: $y = X\beta + \epsilon$
Estimator: $\hat{\beta} = (X^T X)^{-1} X^T y$

6.2 Assumptions (Gauss-Markov)

1. Linearity: y is linear in parameters.
2. Randomness: Samples are random.
3. Non-collinearity: X has full rank.
4. Zero Conditional Mean: $\mathbb{E}[\epsilon|X] = 0$.
5. Homoskedasticity: $\text{Var}(\epsilon|X) = \sigma^2 I$.

6.3 Metrics

- R^2 (Coefficient of Determination): Fraction of variance explained by the model. $R^2 = 1 - \frac{\text{Sum Squared Residuals}}{\text{Total Sum of Squares}}$.
  Warning: Always increases with more variables.
- **Adjusted** R^2: Penalizes complexity (number of regressors). Used to compare models with different numbers of parameters.

- **t-statistic**: Significance of individual coefficients ($\hat{\beta}/SE(\hat{\beta})$). Rules of thumb: $|t| > 2$ is roughly 95% significant.
- **F-statistic**: Significance of the *entire* regression (all $\beta = 0$?).

Practice problems for this section have been moved to `problem_bank/regression/`.

7 Python for Quants

Intuition

Python is the industry standard for research and data analysis. It allows for rapid prototyping and has a rich ecosystem (pandas, numpy). The interview focus is on understanding the language internals (memory, GIL, decorators) and writing idiomatic, vectorized code (avoiding loops). Efficiency in data manipulation is key.

Motivating Example

Vectorization: Calculating the dot product of two arrays of size 10^6.
A naive Python loop takes ∼500 ms, while NumPy's `np.dot(a, b)` takes ∼1 ms. The NumPy version delegates to C/BLAS, running 500x faster. In quant finance, this speed difference is the difference between specific backtests taking 1 hour vs 1 minute.

Key points

- Prefer vectorized NumPy/pandas; avoid Python loops in hot paths.
- Be mindful of copies vs views; use `values` or `to_numpy()` sparingly to avoid hidden copies.
- Use profiling (`%%timeit`, `cProfile`) before optimizing.
- In pandas, align on keys/index to avoid silent misjoins; check dtypes (categorical vs object).

7.1 Data Structures

- **List**: Mutable, ordered.
- **Tuple**: Immutable, ordered.
- **Set**: Mutable, unordered, unique.
- **Dict**: Key-value pairs.

7.2 Key Libraries

- **NumPy**: Array broadcasting, vectorization.
- **Pandas**: Series, DataFrame, 'merge', 'groupby'.

Practice problems for this section have been moved to `problem_bank/compute/`.

8 Finance Concepts

Intuition

Finance is the application of these mathematical tools to markets. We focus on No-Arbitrage (the law of one price), Risk-Return tradeoffs, and Portfolio theory (optimizing the Sharpe ratio). The interview tests your ability to link math to PnL and understand factor exposures.

Key points (intraday equities tilt)

- Always break PnL into factors: alpha, beta exposure, costs (spread/impact/timing), and residual.
- Benchmark choice matters: Arrival for urgency, VWAP/POV for footprint control; execution quality = benchmark vs fills.
- Diversification helps until the correlation floor; covariance quality drives risk estimates—monitor stability.

Practice problems for this section have been moved to `problem_bank/finance/`.

9 Time Series

Intuition

Time-series analysis is the backbone of intraday alpha research. We model how prices and signals evolve, test for stationarity/cointegration, and forecast volatility. The key question is always: does this series have memory we can exploit, or is it a random walk?

Key points

- Time-series: stationarity tests + half-life; cointegration steps; vol forecasts (EWMA/GARCH).
- ACF/PACF heuristics identify AR vs MA structure; Ljung–Box checks residual whiteness.
- Intraday seasonality must be removed before testing for alpha.

9.1 Core Concepts

- **AR(1) Stationarity**: $X_t = \phi X_{t-1} + \epsilon_t$, $|\phi| < 1$. Yule–Walker: $\hat{\phi} = \frac{\sum X_t X_{t-1}}{\sum X_{t-1}^2} = \hat{\rho}_1$.
- **ADF Null**: H_0: unit root. Regression $\Delta X_t = \alpha + \beta t + \gamma X_{t-1} + \sum \psi_i \Delta X_{t-i} + \epsilon_t$. Reject if t_γ more negative than crit (nonstandard).
- **Cointegration Recipe**: Regress Y on X (OLS), take residual e_t. ADF on e_t (no intercept/trend). If stationary, (X, Y) are cointegrated. Use half-life $= \ln(0.5)/\ln(\hat{\phi})$ on residual AR(1) for reversion speed.
- **Vol Forecasting**: EWMA $\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1-\lambda) r_{t-1}^2$ (RiskMetrics $\lambda \approx 0.94$). GARCH(1,1) $\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta \sigma_{t-1}^2$, persistence $\alpha + \beta$.
- **State-Space/Kalman One-Liner**: Latent $x_t = F x_{t-1} + w_t$, observe $y_t = H x_t + v_t$. Kalman provides recursive MMSE; handy for noisy mid/quote filtering.
- **ACF/PACF heuristic**: Slow geometric decay $\to$ AR; sharp cutoff then noise $\to$ MA. Use Ljung–Box to test residual whiteness.
- **Intraday seasonality**: Dampen/standardize by time-of-day vol/mean profiles before testing for alpha; reapply scales when trading.

Practice problems have been moved to `problem_bank/time_series/`.

10 Special Topics

Optimization with costs, microstructure/execution, whiteboard recipes, and interview strategy.

10.1 Optimization with Costs

- **Mean-Var + Turnover (QP)**: $\min_w \frac{1}{2}w^T\Sigma w - \lambda\mu^T w + \kappa\|w - w_0\|_2^2$ s.t. $\mathbf{1}^T w = 1$, box/sector constraints. KKT: $\Sigma w - \lambda\mu + 2\kappa(w - w_0) + \nu\mathbf{1} + A^T\eta = 0$.
- **Woodbury/Sherman–Morrison**: $(A + UCV)^{-1} = A^{-1} - A^{-1}U(C^{-1} + VA^{-1}U)^{-1}VA^{-1}$. Useful for rapid rebalances when Σ updated low-rank.
- **L1 vs L2**: L1 promotes sparsity/turnover control (many zero trades). L2 smooths positions, improves conditioning, no sparsity. Pick based on fill frictions and PM style.
- **Transaction costs**: Linear (spread/fees/impact) vs quadratic (slippage with participation squared). Quadratic cost folds into the Hessian; linear can be handled via L1/prox or piecewise linear solvers.
- **Risk caps**: Add constraints $|w_i| \leq c_i$, factor exposures $F^T w = b$, sector weights. In KKT, they add multipliers tightening the gradient equation.

10.2 Microstructure & Execution

- **Slippage Decomposition**: slippage = spread/fees + impact (temporary/permanent) + timing risk.
- **Benchmark Choice**: Arrival (risk-on, cost sensitive), VWAP (participation neutrality), POV (caps footprint). State when each is superior (thin tape → POV; newsy open → limit arrival risk).
- **Queue/Spread Capture**: price-time priority; joining vs taking; hidden/iceberg; adverse selection if quoting when imbalance/short-term alpha predicts move.
- **Almgren–Chriss intuition**: Cost $\propto kv$ (temp impact), risk $\propto \sigma^2 T$. Higher risk aversion → faster schedule (front-load); lower → flatter POV-like path.
- **Order book stats**: Spread, top-of-book depth, imbalance $= (B - A)/(B + A)$; rising imbalance predicts drift but increases adverse selection risk if you quote.
- **Participation**: Cap child orders to a share of volume to avoid footprint blowout; higher participation → more impact, less timing risk.

10.3 Whiteboard Cheats (full step outlines)

- **AR(1) half-life**:
  1. Model: $X_t = \phi X_{t-1} + \epsilon_t$, $|\phi| < 1$ for stationarity.
  2. Take expectation iteratively: $\mathbb{E}[X_t] = \phi^t X_0$ (decays geometrically).
  3. Define half-life h by $\phi^h = 1/2$.
  4. Solve: $h = \ln(0.5)/\ln(\phi)$.

- **Cointegration test**:
  1. Run OLS: $Y_t = \alpha + \beta X_t + e_t$; save residuals $\hat{e}_t$.
  2. Run ADF on $\hat{e}_t$ with no intercept/trend: $\Delta\hat{e}_t = \gamma\hat{e}_{t-1} + \sum \psi_i \Delta\hat{e}_{t-i} + u_t$.
  3. Check t-stat on γ against critical values (Engle–Granger table).
  4. If reject unit root ($\gamma < 0$ significant), residual is stationary $\Rightarrow (X, Y)$ cointegrated.
  5. Optional: fit AR(1) on $\hat{e}_t$ to get half-life of spread: $\ln(0.5)/\ln(\hat{\phi})$.
- **Ljung–Box on residuals**:
  1. Compute sample autocorr $\hat{\rho}_k$ up to lag m on residuals.
  2. Form $Q = n(n+2)\sum_{k=1}^{m} \hat{\rho}_k^2/(n-k)$.
  3. Compare Q to χ^2_m critical value at chosen α.
  4. If $Q >$ crit: residuals not white $\Rightarrow$ add lags/structure; else pass whiteness check.
- **No-trade band with linear costs**:
  1. Objective: $\max_w \; aw - c|w| - \frac{1}{2}\lambda w^2$ (a=alpha, c=cost per unit, λ=risk penalty).
  2. Case 1: $|a| \le c \Rightarrow$ optimal $w = 0$ (dead zone).
  3. Case 2: $a > c \Rightarrow w^\star = (a-c)/\lambda$.
  4. Case 3: $a < -c \Rightarrow w^\star = (a+c)/\lambda$ (negative).
  5. Intuition: costs shift alpha by c before risk scaling.
- **Woodbury rank-1 inverse update**:
  1. Goal: invert $A + uu^T$ quickly given A^{-1}.
  2. Formula: $(A + uu^T)^{-1} = A^{-1} - \frac{A^{-1}uu^TA^{-1}}{1+u^TA^{-1}u}$.
  3. Steps to write: compute scalar $s = 1 + u^TA^{-1}u$; compute vector $v = A^{-1}u$; update inverse as $A^{-1} - vv^T/s$.
- **KKT (equality constraint example)**:
  1. Problem: $\min_w f(w)$ s.t. $g(w) = 0$.
  2. Lagrangian: $\mathcal{L}(w, \nu) = f(w) + \nu g(w)$.
  3. Stationarity: $\nabla f(w) + \nu \nabla g(w) = 0$.
  4. Primal feasibility: $g(w) = 0$.
  5. Example mean-variance with budget: $f = \frac{1}{2}w^T\Sigma w - \lambda\mu^T w$, $g = \mathbf{1}^T w - 1$.
  6. Stationarity becomes: $\Sigma w - \lambda\mu + \nu\mathbf{1} = 0$; plus $\mathbf{1}^T w = 1$.

10.4 When in Doubt (stuck playbook)

- **Restate + slice**: Rephrase the question back; reduce to the smallest nontrivial variant (2 assets, AR(1), 1 constraint, 1 lag) and solve that. Say you're starting simple, then generalize if time.
- **Declare assumptions**: Say them out loud (stationarity, homoskedasticity, independence, linear impact, no lookahead). Makes partial answers acceptable.

- **Anchor with a baseline**: Write the canonical form to ground you: risk $w^T \Sigma w$, AR(1) $\hat{\phi} = \rho_1$, cost split = spread+impact+timing, regression $\hat{\beta} = (X^T X)^{-1} X^T y$. For execution, always split slippage into spread/impact/timing.
- **Use approximations**: If algebra is messy, give first-order/Taylor or bounds: $\ln(1 + x) \approx x$, drop tiny terms, "if $\lambda \to 0$ you chase alpha; if $\kappa \to \infty$ you hug w_0."
- **Check units/signs/extremes**: Are you mixing bps vs dollars? Variance vs vol? Costs should be positive, no-trade bands symmetric. Sanity-check edge cases ($\phi = 0$ no memory, $\phi = 1$ random walk; $\alpha + \beta < 1$ for GARCH stability).
- **Propose a validation**: If it's a model/alpha question, say: "I'd hold out, walk-forward, check leakage, Ljung–Box residuals, and do an IS/OOS decay plot." Shows you know how to avoid overfit.
- **List options with tradeoffs**: Offer 2–3 choices and pick one: Arrival vs VWAP vs POV; L1 vs L2; add a lag vs keep model parsimonious. Give one-sentence pros/cons.
- **Draw a tiny example**: 2x2 covariance, 3-period time series, 2 boxes in a puzzle. Solve that to show method, then say "same logic scales."
- **Ask one clarifier**: Tight, binary: "Assume independent increments?" "Treat costs linear or quadratic?" "Need exact or approximation?" Avoid broad fishing.
- **Narrate your steps**: Say what you're doing: "I'll start with the simplest case, write the objective, take derivative/KKT, then add costs." Silence kills; narration buys credit.
- **If memory blank, cite principle**: "By Perron–Frobenius, stochastic matrix has $\lambda_{\max} = 1$"; "By CLT, average tends to Normal, so I'll use a Z-approximation"; "By law of large numbers, the sample mean converges to the true mean." Principles score even if you don't finish.
- **Close with a plan**: If still stuck, outline what you'd do next with time/computer: simulate, numerically optimize, or look up a critical value. Shows pragmatism.

Practice problems have been moved to `problem_bank/special_topics/`.